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**Karmaveer Bhaurao Patil College of Engineering,
Satara**

Department: Computer Science & Engineering

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Project Synopsis

On

**“WeatherGPT : Conversational AI for Weather
Forecasting, Alerts, and Climate Information”**

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1.Introduction

Agriculture is highly dependent on weather conditions, making accurate and timely weather information essential for effective farming decisions. Farmers regularly need information about rainfall, temperature, humidity, wind conditions, severe weather, and changing climate patterns to plan activities such as sowing, irrigation, fertilizer application, harvesting, and crop protection. However, conventional weather applications often provide raw forecasts without explaining how the information can be applied to specific agricultural situations.

WeatherGPT: Conversational AI for Smart Weather Forecasting, Alerts and Climate Information for Farmers is proposed as an intelligent, conversational weather assistance system designed to bridge this gap. The system combines weather data, climate information, and Artificial Intelligence to provide farmers with understandable and context-aware responses to weather-related queries.

Instead of requiring users to interpret complex weather reports, WeatherGPT allows farmers to interact with the system using natural language. It can provide location-based weather forecasts, explain upcoming weather conditions, generate relevant alerts, and offer agriculture-oriented insights based on the available weather information. The system is intended to make weather information more accessible, actionable, and easier to understand.

The proposed system uses modern technologies such as **Artificial Intelligence, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), weather APIs, and a web-based interface** to process weather information and generate meaningful responses. By combining real-time weather data with AI-driven conversational capabilities, the project aims to create a practical decision-support tool that can help farmers make better-informed decisions while reducing the difficulty of interpreting weather information.

2. Literature Review

Recent advancements in Artificial Intelligence, Large Language Models (LLMs), and machine learning-based weather forecasting have created new opportunities for developing intelligent agricultural decision-support systems. Researchers have explored AI-based weather prediction, agricultural chatbots, LLM-based recommendations, and the integration of real-time weather and environmental data.

1. **Kuska, M. T., Wahabzada, M., & Paulus, S. (2024)** investigated the potential applications of Large Language Models in crop production. The study highlighted that LLMs can assist farmers by interpreting complex information, supporting decision-making, and translating AI-generated data into understandable recommendations. However, it also identified limitations related to reliability and the need for appropriate agricultural knowledge sources.
2. **Li et al. (2024)** presented a review of foundation models in smart agriculture. The research discussed the potential of large language and foundation models for agricultural applications while identifying challenges such as data requirements, generalization, reliability, and domain-specific adaptation.
3. **Park & Choi (2024)** proposed an LLM-based approach for agricultural meteorological recommendations. Their work used prompt engineering and multiple rounds of refinement to generate more relevant recommendations from weather and agricultural information. The study demonstrated the potential of LLMs to convert meteorological information into actionable agricultural recommendations.
4. **Akilan & Baalamurugan (2024)** developed an automated weather forecasting and field-monitoring system using a GRU-CNN model and IoT technologies for precision agriculture. Their work demonstrates how weather forecasting and field information can be combined to support agricultural decision-making.
5. **Wang & Karimi (2024)** explored the use of Large Language Models for climate forecasting and investigated the capability of GPT-4 to provide rainfall predictions at different time scales. The research identified both the potential of LLMs for making climate information accessible and the limitations of using general-purpose LLMs directly for reliable numerical forecasting.
6. **Zhang et al. (2025)** examined foundation models as assistive tools in hydrometeorology. The study found that foundation models can be useful for data processing, event diagnosis, and decision-support tasks, but highlighted concerns including hallucination, reliability, over-reliance, and the need for domain-specific systems.
7. **Kow et al. (2025)** proposed an AI-driven weather downscaling approach using convolutional autoencoders and LSTM-Transformer networks for smart agriculture. Their results showed improvements in temperature, humidity, vapor pressure deficit, and heat-index forecasting, demonstrating how improved weather information can support agricultural management and crop protection.
8. **Park et al. / CottonBot (2025)** presented an AI-powered agricultural assistant combining LLMs, Retrieval-Augmented Generation (RAG), weather forecast APIs, and IoT soil-moisture sensors. The system demonstrates the advantage of combining static agricultural knowledge with real-time environmental data to generate field-specific recommendations.

9. **AgroBuddy (2025)** proposed an AI-driven chatbot for Indian farmers using LLaMA-3, RAG, agricultural information sources, market information, and weather forecasts. The work demonstrates the feasibility of multilingual, data-driven agricultural assistance for Indian farming communities.
10. **Krzyżewska (2025)** reviewed the application of AI tools in weather and climate-related tasks and evaluated several contemporary AI systems. The study highlighted the growing role of generative AI in interpreting meteorological information while also demonstrating limitations in tasks requiring accurate retrieval and interpretation of historical weather data.

Research Gap

The literature indicates that AI can significantly improve weather forecasting, agricultural recommendations, and accessibility of meteorological information. However, many existing systems focus either on **weather prediction** or on **general agricultural advisory services**. General-purpose LLMs also cannot be relied upon as standalone numerical weather forecasting systems because of their limitations in producing accurate real-time meteorological predictions. Therefore, the proposed **WeatherGPT** focuses on integrating **real-time weather APIs, agricultural knowledge, LLM-based natural-language interaction, RAG, and an alert mechanism** into a single farmer-oriented platform. Rather than attempting to replace professional numerical weather prediction systems, the proposed system aims to act as an intelligent interface that converts reliable weather information into understandable and agriculture-oriented insights for farmers.

Literature Review Milestone

A milestone will be included in the project schedule for the **identification, study, and submission of 8–10 recent standard research papers**, preferably published during **2024–2025**, covering the following areas:

- AI/ML-based weather forecasting
- LLMs and agricultural applications
- RAG-based agricultural assistants
- AI-based meteorological recommendations
- Weather-based agricultural decision support
- AI applications in climate and weather information systems
- Farmer-oriented conversational AI

III. Existing Systems

Several weather and agricultural advisory systems are currently available to provide farmers with weather forecasts, warnings, and agricultural information. However, these systems generally present information through predefined dashboards, notifications, or advisories rather than through an intelligent conversational interface.

3.1 MAUSAM

The **MAUSAM application**, developed by the India Meteorological Department (IMD), provides weather observations, forecasts, radar information, and warnings for different locations. It is primarily focused on general weather information and severe-weather warnings.

3.2 Meghdoot

Meghdoot is an agricultural weather application developed through collaboration between IMD, IITM, and ICAR. It provides location-specific weather forecasts and crop- and district-level agro-meteorological advisories. It also provides information such as rainfall, temperature, humidity, wind conditions, and weather warnings.

3.3 Damini

Damini, developed by IITM, focuses specifically on lightning detection and warnings. It is useful for providing farmers and other users with alerts regarding nearby lightning activity and potential risks.

3.4 Commercial Weather Platforms and APIs

Platforms such as **OpenWeather** provide developers with current weather conditions, forecasts, historical weather information, and weather alerts through APIs. Such services provide the underlying weather data but generally do not themselves provide a farmer-specific conversational AI layer for interpreting the information.

Limitations of Existing Systems

Although existing systems provide valuable and reliable weather information, there are some limitations:

- Information is often presented as **charts, dashboards, forecasts, or predefined advisories**.
- Farmers may need to interpret weather parameters themselves.
- Interaction is generally not **conversational or natural-language based**.
- Agricultural recommendations may not be dynamically generated for every individual query.
- General-purpose weather platforms are not specifically designed around **farmer-oriented decision support**.

The proposed **WeatherGPT** aims to complement these existing systems rather than replace official weather forecasting services. It will use reliable weather data as an input and provide a conversational AI layer capable of explaining weather conditions and generating understandable, agriculture-oriented insights. This approach combines **real-time weather information**,

agricultural knowledge, RAG, and LLM-based natural-language interaction in a unified system.

IV) Proposed Work

a. Problem Statement

Farmers depend on weather conditions for various agricultural activities such as sowing, irrigation, spraying, fertilization, and harvesting. Although existing weather applications provide forecasts, warnings, and meteorological data, the information is often presented in the form of numerical values, charts, or predefined advisories, making it difficult for farmers to interpret and apply it to their specific agricultural needs.

General-purpose AI assistants can answer weather-related questions, but they may not have access to reliable real-time weather data or verified agricultural knowledge. Therefore, there is a need for an intelligent system that can combine **real-time weather information with agricultural knowledge** and present the information through a simple conversational interface.

The proposed **WeatherGPT** is an AI-powered, farmer-focused conversational system that integrates real-time weather data, agricultural knowledge, Retrieval-Augmented Generation (RAG), and Large Language Models (LLMs) to provide understandable weather information, agriculture-oriented insights, and weather alerts.

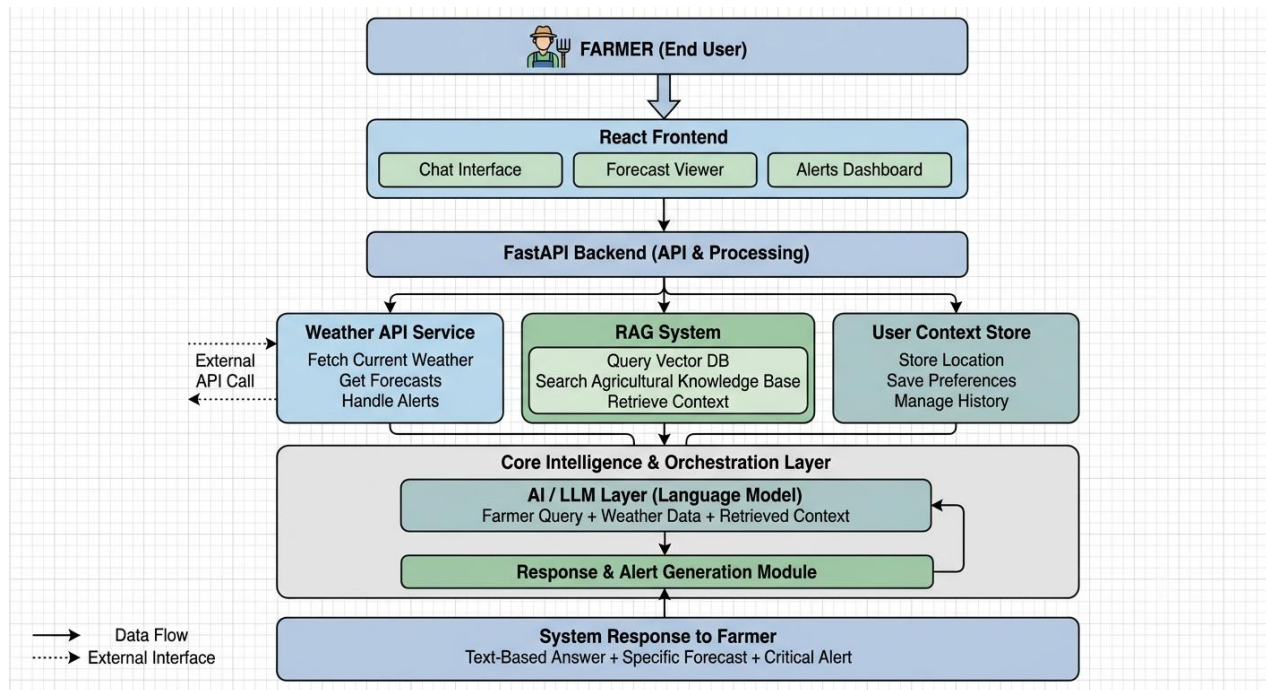
b. Novelty

The proposed system provides the following novel features:

- **Farmer-focused conversational interface** for asking weather and agriculture-related questions in natural language.
- **Integration of real-time weather APIs with an LLM** to provide responses based on current weather conditions.
- **Retrieval-Augmented Generation (RAG)** to retrieve relevant information from a domain-specific agricultural knowledge base.
- **Weather-to-agriculture interpretation**, where weather conditions are explained in terms of their possible relevance to farming activities.
- **Location-aware responses** based on the farmer's selected location.
- **Weather-based alerts** for potentially important conditions such as heavy rainfall, extreme temperatures, strong winds, and storms.
- **Unified platform** combining weather forecasts, alerts, agricultural information, and conversational AI.

The system does not attempt to replace professional numerical weather forecasting services. Instead, it acts as an **AI-based interface and decision-support system** that converts reliable weather information into understandable, farmer-oriented information.

c. Architecture (Block Diagram)



Architecture Description

The **React frontend** provides the interface through which farmers can enter natural-language queries, select their location, view forecasts, and receive alerts.

The **FastAPI backend** manages communication between the frontend, weather services, database, RAG system, and LLM.

The **Weather API** supplies real-time weather conditions, forecasts, and available weather alerts.

The **RAG system** retrieves relevant information from the agricultural knowledge base according to the user's query.

The retrieved agricultural information and real-time weather data are provided to the **LLM**, which generates a contextual and easy-to-understand response. The final response and relevant alerts are then displayed to the farmer through the frontend.

d. Modules

1. User Management Module

- User registration and login
- User profile management
- Location selection
- User preferences

2. Weather Data Module

- Current weather conditions
- Temperature and humidity
- Rainfall and precipitation
- Wind speed and direction
- Weather forecasts
- Severe weather information

3. Agricultural Knowledge Base Module

- Crop-related information
- Irrigation practices
- Sowing and harvesting information
- Weather-related farming practices
- Pest and disease information
- Verified agricultural resources

4. RAG Module

- Processes the user's query
- Searches the agricultural knowledge base
- Retrieves relevant information
- Provides retrieved context to the LLM
- Reduces dependence on the LLM's internal knowledge

5. Conversational AI / LLM Module

- Natural-language query understanding
- Context-aware response generation
- Integration of weather data and agricultural knowledge
- Conversational interaction with farmers
- Simplification of complex weather information

6. Weather Alert Module

- Detection of significant weather conditions
- Heavy rainfall alerts
- Extreme temperature alerts
- Strong wind/storm alerts
- Presentation of alerts in simple language

7. Agricultural Decision-Support Module

- Interprets weather conditions from an agricultural perspective
- Provides weather-related farming insights
- Helps farmers understand possible effects on agricultural activities
- Provides contextual information based on available data

8. Database Module

- User information
- Locations
- Conversation history
- Alert history
- System records

9. Backend API Module

- Handles frontend requests

- Connects external weather APIs
- Communicates with the RAG and LLM components
- Manages database operations
- Controls application logic

V) Project Objectives:

1. To learn Generative AI and Large Language Models (LLMs) and understand how they can be integrated into real-world applications.
2. To learn Retrieval-Augmented Generation (RAG) for retrieving relevant information from a domain-specific knowledge base and providing contextual responses.
3. To learn Prompt Engineering and LLM integration for designing effective prompts and controlling AI-generated responses.
4. To learn Python-based backend development using FastAPI for designing REST APIs and connecting different components of the system.
5. To learn Vector Databases and Embeddings for implementing semantic search and storing/retrieving agricultural knowledge for the RAG pipeline.
6. To learn integration of third-party APIs, particularly weather APIs, to obtain and process real-time meteorological data.
7. To strengthen React.js skills by developing an interactive web interface for conversational AI, weather forecasts, dashboards, and alerts.
8. To learn database management using PostgreSQL for storing user information, locations, conversations, and application data.
9. To learn Docker and containerization for creating reproducible development and deployment environments.
10. To understand AI application architecture and system integration, including communication between the frontend, backend, external APIs, databases, RAG pipeline, and LLM services.
11. To gain practical knowledge of AI system evaluation, including response relevance, retrieval quality, accuracy, latency, and usability.

Technologies to be Learned

Core: Python, FastAPI, REST APIs

AI: LLMs, Generative AI, Prompt Engineering, RAG

AI Infrastructure: Embeddings, Vector Database, Semantic Search

Frontend: React.js

Database: PostgreSQL

External Services: Weather APIs, LLM APIs

DevOps: Docker, Git/GitHub

The main objectives of the proposed **WeatherGPT** system are:

1. **To develop a farmer-focused conversational AI system** that enables users to interact with weather and agricultural information using natural language.
2. **To provide real-time, location-specific weather information** such as temperature, rainfall, humidity, wind conditions, forecasts, and weather alerts.
3. **To integrate Large Language Models (LLMs) with real-time weather data** to generate meaningful and context-aware responses.
4. **To implement Retrieval-Augmented Generation (RAG)** using a domain-specific agricultural knowledge base to provide relevant and reliable agricultural information.
5. **To convert complex weather information into simple, understandable insights** that can help farmers interpret its possible impact on agricultural activities.
6. **To provide timely weather-based alerts** for potentially significant conditions such as heavy rainfall, extreme temperatures, strong winds, and storms.
7. **To develop a unified web-based platform** that combines weather forecasting, agricultural information, alerts, and conversational AI in a single interface.
8. **To improve accessibility of weather and agricultural information** by allowing users to obtain relevant information through simple natural-language queries.
9. **To evaluate the system based on response relevance, information retrieval accuracy, response time, and usability** for farmer-oriented weather and agricultural queries.

Academic Objectives

The academic objectives of the proposed project are to gain practical knowledge and hands-on experience in the following emerging technologies:

1. **To learn Generative AI and Large Language Models (LLMs)** and understand how they can be integrated into real-world applications.
2. **To learn Retrieval-Augmented Generation (RAG)** for retrieving relevant information from a domain-specific knowledge base and providing contextual responses.
3. **To learn Prompt Engineering and LLM integration** for designing effective prompts and controlling AI-generated responses.
4. **To learn Python-based backend development using FastAPI** for designing REST APIs and connecting different components of the system.
5. **To learn Vector Databases and Embeddings** for implementing semantic search and storing/retrieving agricultural knowledge for the RAG pipeline.
6. **To learn integration of third-party APIs**, particularly weather APIs, to obtain and process real-time meteorological data.
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Technologies to be Learned

Core: Python, FastAPI, REST APIs

AI: LLMs, Generative AI, Prompt Engineering, RAG

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Frontend: React.js

Database: PostgreSQL

External Services: Weather APIs, LLM APIs

DevOps: Docker, Git/GitHub

Requirement Analysis

a. Software Requirements

1. Operating System: Windows 10/11 or Ubuntu Linux 22.04 or later
2. Programming Language: Python 3.10+, JavaScript
3. Frontend Technology: React.js, HTML5, CSS3
4. Backend Technology: FastAPI
5. Database: PostgreSQL
6. AI Technologies: Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG), Prompt Engineering and Embeddings
7. Vector Database: pgvector
8. Weather Service: Weather API for real-time weather and forecast data
9. Development Environment: Visual Studio Code
10. API Testing Tool: Postman
11. Version Control: Git and GitHub
12. Containerization: Docker and Docker Compose
13. Web Browser: Google Chrome / Mozilla Firefox / Microsoft Edge
14. Internet Connection: Required for accessing weather APIs and cloud-based AI services

b. Hardware Requirements

1. Processor: Intel Core i5 or equivalent AMD processor and above
2. RAM: Minimum 8 GB, preferably 16 GB
3. Storage: Minimum 10 GB free storage space
4. Hard Disk/SSD: SSD recommended
5. Display: Minimum 1366 × 768 resolution
6. Keyboard and Mouse
7. Stable Internet Connection
8. GPU: Not required when using cloud-based LLM APIs

c. Operating System Requirement

The proposed system can be developed and executed on Windows 10/11 or Ubuntu Linux 22.04 and later versions. Windows 11 with WSL2 and Docker or Ubuntu Linux is recommended for development and deployment. The system requires a modern web browser and a stable Internet connection for accessing external weather APIs and cloud-based AI/LLM services.

Project Management Plan

a. Gantt Chart

Activities	W1- W2	W3- W4	W5- W6	W7- W8	W9- W10	W11- W12	W13- W14	W15- W16
Problem Identification & Requirement Analysis	✓							
Literature Survey	✓	✓						
System Design & Architecture		✓						
Learning Required Technologies		✓	✓					
Frontend Development			✓	✓				
Backend Development			✓	✓	✓			
Weather API Integration				✓				
Database Development				✓	✓			
RAG & Knowledge Base					✓	✓		
LLM Integration					✓	✓		
Alert & Decision-Support Module						✓		
System Integration & Testing						✓	✓	
Performance & Usability Testing							✓	
Documentation & Final Report							✓	✓
Final Presentation & Demonstration								✓

b. Weekly Milestones

Week	Milestone
Week 1	Finalize project problem, scope and objectives
Week 2	Complete requirement analysis and initial literature survey
Week 3	Study 8–10 recent research papers and identify research gap
Week 4	Finalize system architecture, technology stack and database design
Week 5	Learn required technologies: React, FastAPI, APIs, PostgreSQL and Git
Week 6	Develop initial React frontend and user interface
Week 7	Implement backend APIs and database connectivity
Week 8	Integrate weather API and implement location-based weather retrieval
Week 9	Prepare agricultural knowledge base and implement embeddings/vector storage
Week 10	Develop and test the RAG pipeline
Week 11	Integrate LLM with weather data and RAG pipeline
Week 12	Implement conversational AI and contextual response generation

Week Milestone

Week 13 Develop weather alerts and agriculture-oriented decision-support features

Week 14 Integrate all modules and perform functional testing

Week 15 Perform performance, usability and response-quality testing; fix errors

Week 16 Complete documentation, final demonstration and project presentation

c. Risk Analysis

Risk	Probability	Impact	Mitigation
Weather API failure or unavailability	Medium	High	Use error handling and an alternative weather API/fallback data
LLM API failure or service limitations	Medium	High	Implement API error handling and maintain modular LLM integration
Incorrect or hallucinated AI responses	Medium	High	Use RAG, verified knowledge sources, prompt constraints and response validation
Poor quality of retrieved information	Medium	Medium	Use reliable agricultural sources and improve document chunking and retrieval
Database connectivity issues	Low	Medium	Perform regular backups and use proper database configuration
Integration problems between frontend and backend	Medium	Medium	Test APIs independently using Postman before integration
Insufficient hardware resources	Low	Medium	Use cloud-based LLM services and lightweight local components
Delay in development	Medium	High	Follow weekly milestones and prioritize core features
Security and privacy issues	Low	High	Implement authentication, input validation and secure API key management
Lack of sufficient training data	Medium	Medium	Use reliable public agricultural and meteorological information sources

d. Work Distribution

The project work will be distributed among the team members according to their respective responsibilities.

Team Member	Primary Responsibilities
Abhimanyu R. Gaikwad	Project coordination, system architecture, backend development, LLM/RAG integration, API integration and database management
Shivaji A. Herkal	Frontend development, React UI, dashboard, chatbot interface and user experience
Mayuresh B. Bhosale	Agricultural knowledge-base preparation, literature survey, RAG data processing and testing
Vivek Shinde	Weather alert module, testing, performance evaluation, documentation and deployment support

All team members will participate in **requirement analysis, research-paper study, system integration, testing, documentation, and final presentation.**

Advantages of the Proposed System

1. **Farmer-Friendly Interface:** Provides weather and agricultural information through a simple conversational interface using natural-language queries.
2. **Real-Time Weather Information:** Provides current weather conditions, forecasts, and available weather alerts using external weather APIs.
3. **Location-Based Information:** Provides weather information relevant to the user's selected geographical location.
4. **AI-Powered Assistance:** Uses Large Language Models (LLMs) to understand user queries and generate contextual responses.
5. **Agricultural Knowledge Integration:** Uses a domain-specific agricultural knowledge base to provide relevant farming-related information.
6. **Improved Response Reliability:** Retrieval-Augmented Generation (RAG) allows the system to retrieve relevant information from external knowledge sources before generating responses.
7. **Weather-Based Agricultural Insights:** Helps users understand how weather conditions may affect activities such as irrigation, sowing, spraying, and harvesting.
8. **Timely Weather Alerts:** Provides notifications or warnings for potentially significant weather conditions such as heavy rainfall, extreme temperatures, and strong winds.
9. **Unified Platform:** Combines weather forecasts, agricultural information, AI-based conversation, and alerts within a single system.
10. **Scalable Architecture:** The modular architecture allows additional crops, agricultural knowledge sources, weather services, languages, and AI capabilities to be added in the future.
11. **Reduced Complexity:** Converts complex weather parameters and technical information into simpler and more understandable responses.
12. **Accessible Through Web Interface:** The system can be accessed through a web browser without requiring specialized hardware or software.

Limitations of the Proposed System

1. **Dependence on External Weather APIs:** The accuracy and availability of weather information depend on the reliability, coverage, and limitations of the selected weather API.
2. **Internet Dependency:** Real-time weather data and cloud-based AI services require a stable Internet connection. The functionality may be limited when Internet connectivity is unavailable.
3. **LLM Limitations:** AI-generated responses may occasionally contain incorrect or incomplete information. The system therefore cannot guarantee completely error-free responses.
4. **RAG Knowledge Limitations:** The quality of agricultural responses depends on the accuracy, reliability, and coverage of the documents available in the knowledge base.
5. **Limited Weather Prediction Capability:** The proposed system uses external weather forecasting services and does not independently develop a numerical weather prediction model. Therefore, it cannot guarantee more accurate forecasts than the underlying weather service.
6. **API and Service Costs:** Extensive use of commercial weather APIs, LLM APIs, or other cloud services may involve usage limits or additional costs.
7. **Language Limitations:** Initial implementation may support only a limited number of languages. Support for additional regional languages may require further development and suitable language resources.
8. **Agricultural Recommendation Limitations:** The system provides AI-based decision support and general agricultural insights. Its recommendations should not be considered a substitute for professional agricultural, meteorological, or government advice.
9. **Location and Data Limitations:** The usefulness of location-specific recommendations depends on the availability and accuracy of weather and agricultural data for the selected region.
10. **Security and Privacy Risks:** User information, location data, API credentials, and conversation history must be properly secured to prevent unauthorized access.

Future Scope

1. **Multilingual Support:** The system can be extended to support regional Indian languages such as Marathi, Hindi, Gujarati, Kannada, and other local languages to improve accessibility for farmers.
2. **Voice-Based Interaction:** A speech-to-text and text-to-speech system can be integrated to allow farmers to interact with WeatherGPT using voice commands instead of typing.
3. **Crop-Specific Recommendations:** The system can be enhanced to provide recommendations based on specific crops, soil conditions, growth stages, and farming practices.
4. **IoT Sensor Integration:** Sensors for soil moisture, temperature, humidity, and other environmental parameters can be integrated to provide real-time field-level information.
5. **Advanced Weather Forecasting:** Machine learning and deep learning models can be incorporated in the future to develop localized weather prediction and forecasting capabilities.
6. **Satellite and Remote-Sensing Data:** Satellite imagery and remote-sensing data can be integrated for crop monitoring, drought assessment, vegetation analysis, and weather-impact assessment.
7. **Personalized Agricultural Assistance:** The system can maintain user preferences, crop details, location, and historical information to provide more personalized agricultural insights.
8. **Early Warning and Notification System:** A dedicated notification mechanism can be developed to deliver important weather warnings through mobile notifications, SMS, or other communication channels.
9. **Integration with Government Services:** The system can be integrated with reliable government agricultural and meteorological services to provide official advisories, schemes, alerts, and other relevant information.
10. **Mobile Application:** A dedicated Android/iOS application can be developed to make the system more accessible to farmers.
11. **Advanced AI Agents:** Multiple specialized AI agents can be introduced for weather analysis, crop advisory, pest and disease information, and agricultural planning.
12. **Scalable Deployment:** The system can be deployed on cloud infrastructure to support a larger number of users and enable continuous updates to we

References

- [1] M. T. Kuska, M. Wahabzada, and S. Paulus, “Large Language Models in Agriculture: Opportunities and Challenges for Crop Production,” *Computers and Electronics in Agriculture*, 2024.
- [2] Li et al., “Foundation Models in Smart Agriculture: Applications, Challenges and Opportunities,” *Computers and Electronics in Agriculture*, 2024.
- [3] S. Park and J. Choi, “Large Language Models for Enhanced Agricultural Meteorological Recommendations,” arXiv preprint arXiv:2408.04640, 2024.
- [4] Akilan and Baalamurugan, “Automated Weather Forecasting and Field Monitoring System for Precision Agriculture Using GRU-CNN and IoT,” *Expert Systems with Applications*, 2024.
- [5] Wang and Karimi, “Exploring Large Language Models for Climate Forecasting,” arXiv preprint, 2024.
- [6] Zhang et al., “Foundation Models as Assistive Tools in Hydrometeorology,” *Water Resources Research*, 2025.
- [7] Kow et al., “AI-Driven Weather Downscaling for Smart Agriculture Using Deep Learning,” *Computers and Electronics in Agriculture*, 2025.
- [8] “CottonBot: An AI-Powered Agricultural Assistant Using Large Language Models, Retrieval-Augmented Generation and IoT,” *Agricultural Technology*, 2025.
- [9] “AgroBuddy: An AI-Driven Conversational Assistant for Indian Farmers,” *IEEE Conference Proceedings*, 2025.
- [10] Krzyżewska, “Artificial Intelligence Applications in Weather and Climate,” *Atmosphere*, vol. 16, no. 5, 2025.
- [11] India Meteorological Department (IMD), “Gramin Krishi Mausam Sewa (GKMS) and Agricultural Meteorological Services,” Government of India.
- [12] Ministry of Earth Sciences, Government of India, “Weather and Climate Services for Agriculture,” Government of India.
- [13] OpenWeather, “Weather API Documentation,” OpenWeather.
- [14] FastAPI, “FastAPI Documentation,” FastAPI.
- [15] React, “React Documentation,” Meta Open Source.

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